On Academic Writing

(Academese and all that)

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What this is about

- Purpose
- Style
- Thinking
- The human element

What this is not about

- English grammar
- Software/platform usage

“At Cornell University, my professor of European literature, Vladimir Nabokov, changed the way I read and the way I write. Words could paint pictures, I learned from him. Choosing the right word, and the right word order, he illustrated, could make an enormous difference in conveying an image or an idea.”

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- Ruth Bader Ginsburg

What is writing?

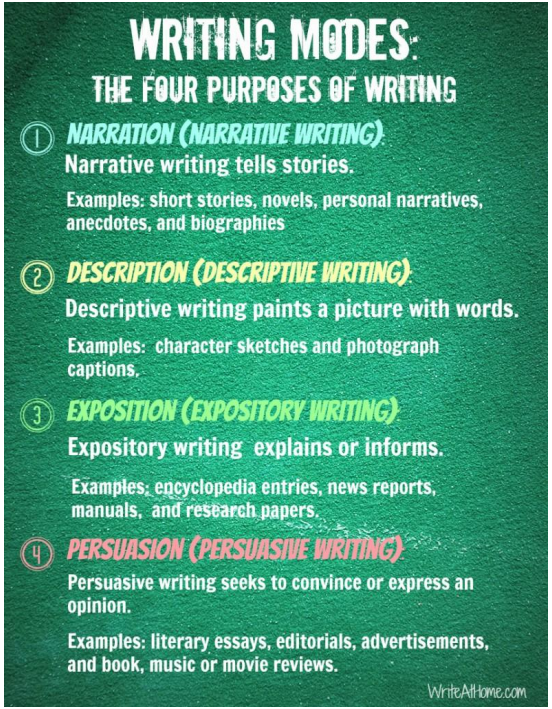
“Man has an instinctive tendency to speak, as we see in the babble of our young children, whereas no child has an instinctive tendency to bake, brew, or write.”

- Charles Darwin

- Conversation is a two-way process -- speaking & listening (more intuitive)
- Writing -- no interlocutor, no live audience
  - Ability to put ourselves in another’s shoes
  - Clear perception of the fantasy world we inhabit in our heads
  - Planning

The purposes of writing

- Brian Wasko of *WriteAtHome.com* identifies four modes of writing:



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- Persuade -- to argue in favour of a theory.
    - e.g. discussion of your interpretation of results, observing/grant proposals, ...

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- Since writing is not something that we do naturally, we may not have a good enough style to fall back on.
- The *classic style* is specifically suited for such situations and is especially helpful for our purposes.

Classic style

- `Clear and Simple as the Truth: Writing Classic Prose`

  Francis-Noël Thomas and Mark Turner

- Focuses on clarity and simplicity and getting the message across
- A style of presentation that endeavours to make itself invisible
- Treats the reader as an equal
- Assumes that the reader is intelligent but does not yet possess the information being conveyed
- Does not hesitate to adopt a conversational tone

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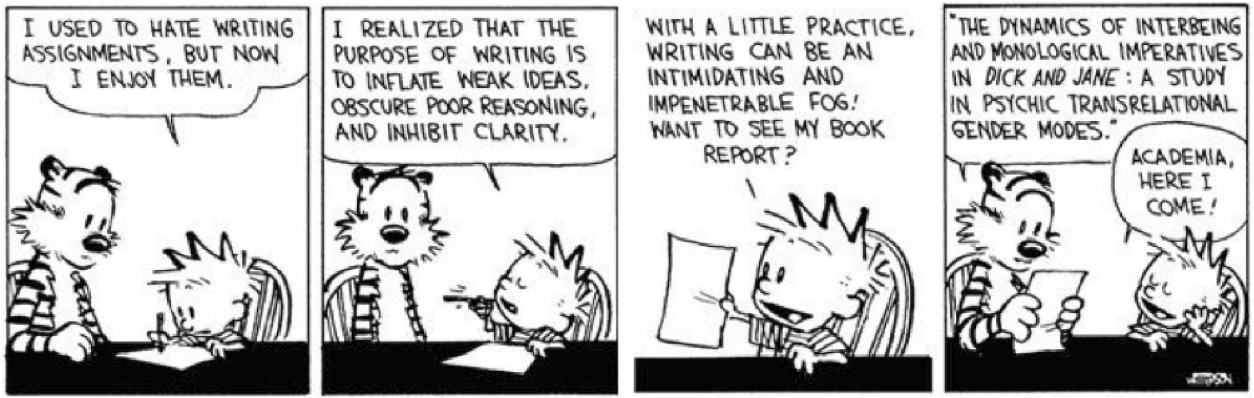

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(https://www.gocomics.com/calvinandhobbes/1993/02/11)

- 1) Self-conscious, long-winded, stuffy prose:
  - I repeatedly come across this when proofreading theses, papers, and research proposals.
  - Longer thesis ≠ Better thesis
  - Fancier words ≠ More precise explanation
    - e.g. “in contradistinction” ⇒ “in contrast”
  - Use action words: e.g. “analysed” ⇒ “performed the analysis”
  - Do not sacrifice clarity and precision in favour of increasing the number of pages or the length of a sentence.
  - Simplicity is good. It is not at all easy to achieve parsimony without practice
  - Worth exploring what makes us use long-winded prose...

- Daniel Oppenheimer, Princeton University:

  “Most texts on writing style encourage authors to avoid overly-complex words. However, a majority of undergraduates admit to deliberately increasing the complexity of their vocabulary so as to give the impression of intelligence …

  … people are more likely to use big words when they are feeling the most insecure. One can imagine that a minority student under stereotype threat (Steele, 1997) might be inclined to increase complexity in his/her writing which would backfire and cause teachers to have lower opinions of the student’s intelligence. Likewise leaders facing crucial decisions might use more complex vocabulary and end up undermining others’ confidence in their leadership ability …

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- He titled this article ***Consequences of erudite vernacular utilized irrespective of necessity: problems with using long words needlessly***.

- 2) Obsession with passive voice:
  - Opting for the passive voice does not automatically make a sentence more objective
    - e.g.*"... spectra were obtained... spectra were extracted... multiple setups were observed… redshifts were determined… frequency was calibrated…"*
  - Using the passive voice unnecessarily can make it seem like the author is trying to evade responsibility, even if they aren't (hint: politicians *love* using the passive voice).

- In *The Sense of Style*, Steven Pinker provides an explanation for why people subconsciously end up writing in the passive:
  - Good writers narrate a story
  - Bad writers work backwards from outcome to cause, writing in the order in which ideas occur to them (which need not be the same as the logical development of ideas)
  - Passive voice makes this easy

- Not a recommendation against using the passive voice -- it depends on what you want to draw the reader's attention to.

- NB: Being a bad writer is NOT the same as being a bad scientist (and has ***nothing to do with being a native speaker*** of the language one writes in). It just requires training.

- 3) Knowing your audience (using professional jargon):
  - Not always deliberate (many well-accomplished academics fall prey to this).
  - This assumes that the reader's state of knowledge is the same as that of the writer.
  - Classic style recommends that the writer assume that the reader's *intelligence* is on par with that of the writer, not their *state of knowledge*.
  - Do not write your thesis as if the intended audience is your advisor or thesis examiner -- the correct audience to aim for is a fellow student in the same field, but not the same subfield.

- 3) Knowing your audience:
  - The twin purposes of classic style are
    - to orient the reader's gaze so that the reader can see for themselves what the writer already sees.
    - engage the reader in a conversation.
  - Count on the reader to be a willing participant. The current sentence must follow logically from the previous one (narrative structure).
  - But no need to spell out everything though.

"If you imagine your 'audience', your readers, imagine them as intelligent and sympathetic — ready to read you if you give them the chance."

- Ursula K. Le Guin

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- 6) Hedging
  - e.g. very, mostly, seemingly, barely, ...
  - Hedging is sometimes unavoidable, but try to minimise it.

Two examples of classic style

- Two examples from two different academic disciplines
  - A sociology / history book
  - A peer-reviewed astronomy paper

Example 1

“Th[e] new reality [of the early 20th century Jim Crow laws] forced colored parents to search for ways to explain the insanity of the caste system to their uncomprehending children. When two little girls in 1930s Florida wanted to know why they couldn’t play on a swing like the white children or had to sit in a dirty waiting room instead of the clean one, their father had to think about how to make them understand. [...]

All told, these statutes only served to worsen race relations, alienating one group from the other and removing the few informal interactions that might have helped both sides see the potential good and humanity in the other.”

- Isabel Wilkerson, *The Warmth of Other Suns (2010)*

Example 2

“[It is] quite fortunate that the emergence of the RIME formalism has provided us with a complete and elegant mathematical framework for dealing with observational effects [...] Oddly enough, outside of a small community of algorithm developers[,] uptake of RIME by radio astronomers at large has been slow… [f]rom my personal experience of teaching the RIME at several workshops, once the penny drops, people tend to describe it in terms such as “obvious”, “simple”, “intuitive”, “elegant” and “powerful”. This points at an explanatory gap in the literature. Paper I of this series therefore tries to address this gap, recasting existing ideas into one consistent mathematical framework, and showing where other approaches to the RIME fit in.”

- Oleg Smirnov, *Revisiting the RIME, Paper I, A&A (2011)*

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  - It’s not easy to write an entire thesis in the last few weeks of your Masters/PhD
- Even if your writing skills are top-notch, starting to write early is necessary to
  - Develop a vision for your project
  - Avoid unpleasant surprises

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- Writer's block
  - Stream of consciousness writing -- "morning pages" by the author Julia Cameron
- Corrections:
  - At first, it can be intimidating when you see lots of corrections from your advisor(s)
    - Moreover, different people WILL give different corrections/opinions on your writing
  - Not an indictment on your language skills; best thought of as suggestions for improvement from people who took the time to read what you wrote, because, that's what it is (mostly!).

What if English is not your first language?

- Writing well vs knowing English well ⇔ Ability to code vs knowing Python well

- You can acquire knowledge of English (Python) over time, and always look grammar (syntax) up online, but learning to write (code) well is separate from this.

- I use the following while writing (especially when writing a thesis or during the later stages of writing a paper)
  - https://books.google.com/ngrams
  - An online thesaurus/dictionary

What if English is not your first language?

- There are advantages to being a second language writer
  - New ways to conceptualise and express ideas available without additional effort
  - Many writers of beautiful English prose were/are exophonic (i.e. English is not their first language)
    - Vladimir Nabokov
    - Elif Shafak
    - Salman Rushdie
    - Kazuo Ishiguro
    - Chinua Achebe

One last thing...

Once in a while, remember to flout any of the above rules judiciously 😊